

Softmax Regression:

cgpa	iq	Y
7.1	71	0
8.5	85	1
9.5	95	2

Yes $\rightarrow 1$
 No $\rightarrow 0$
 opt/out $\rightarrow 2$
 (3 classes)

not binary.

Softmax multinomial LR

Softmax fn:

$$\sigma(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$

$K \rightarrow$ no of classes

(Yes):

$$\sigma(z_1) = \frac{e^{z_1}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

NO 1

$$\sigma(z_2) = \frac{e^{z_2}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

opt/out.

$$\sigma(z_3) = \frac{e^{z_3}}{e^{z_1} + e^{z_2} + e^{z_3}}$$

$0 < \sigma < 1$ (Probability of each class)

$$\sigma_{z_1} + \sigma_{z_2} + \sigma_{z_3} = 1$$

• cgpa

iq

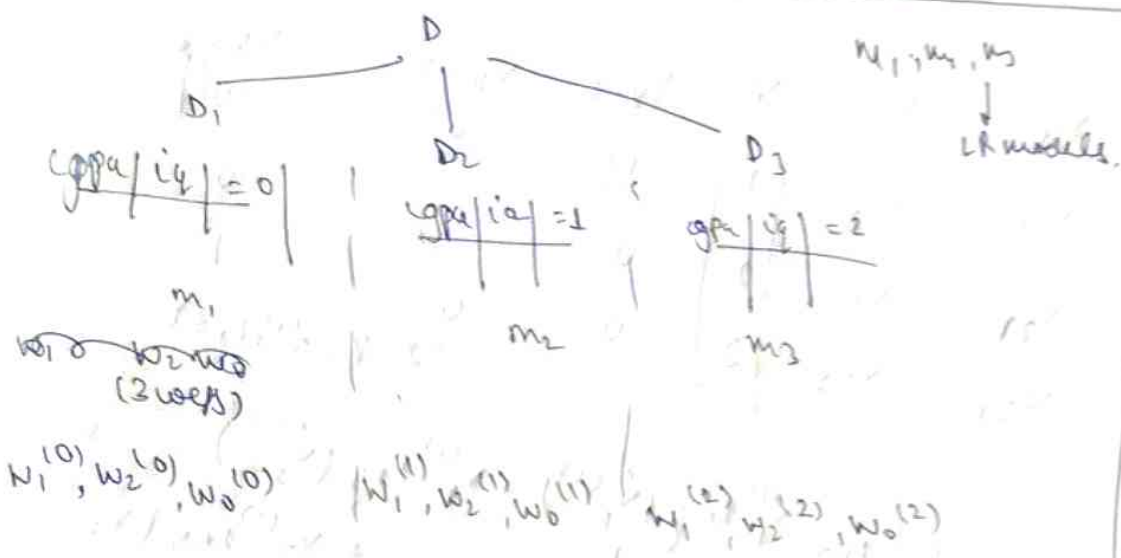
Y

0
1
2

→ transform (ONE)

cgpa	iq	0	1	2
-	-	1	0	0
-	-	0	1	0
-	-	0	0	1

50%



$$S_x = \{7, 70\} ?$$

$m_1(Y)$	$m_2(N)$	$m_3(0)$
$w_1^{(1)}, w_2^{(1)}, w_3^{(1)}$	$w_1^{(2)}, w_2^{(2)}, w_3^{(2)}$	$w_1^{(3)}, w_2^{(3)}, w_3^{(3)}$
$z_1 = 7 \times w_1^{(1)} + 70 \times w_2^{(1)} + w_3^{(1)}$	$z_2 = 7 \times w_1^{(2)} + 70 \times w_2^{(2)} + w_3^{(2)}$	$z_3 = 7 \times w_1^{(3)} + 70 \times w_2^{(3)} + w_3^{(3)}$

$$\sigma(Y) = 0.4$$

$$\sigma(N) = 0.35$$

$$\sigma(0) = 0.25$$

as $\sigma(Y)$ is >, parent node.

for large classes, many 'm' makes slow:

logitⁿ:

$$L = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1-y_i) \log(1-\hat{y}_i)]$$

logistic Regression.

logitⁿ: (softmax):

$$L = -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K y_k^{(i)} \log(\hat{y}_k^{(i)})$$

x_1	x_2	Y	$Y_{k=1}$	$Y_{k=2}$	$Y_{k=3}$
x_{11}	x_{12}	1	1	0	0
x_{21}	x_{22}	2	0	1	0
x_{31}	x_{32}	3	0	0	1

internal part:

$$\begin{aligned}
 & Y_1^{(1)} \log(\hat{Y}_1^{(1)}) + Y_2^{(1)} \log(\hat{Y}_2^{(1)}) + Y_3^{(1)} \log(\hat{Y}_3^{(1)}) \\
 & + Y_1^{(2)} \log(\hat{Y}_1^{(2)}) + Y_2^{(2)} \log(\hat{Y}_2^{(2)}) + Y_3^{(2)} \log(\hat{Y}_3^{(2)}) \\
 & + Y_1^{(3)} \log(\hat{Y}_1^{(3)}) + Y_2^{(3)} \log(\hat{Y}_2^{(3)}) + Y_3^{(3)} \log(\hat{Y}_3^{(3)})
 \end{aligned}$$

$$L = -\frac{1}{n} \left[Y_1^{(1)} \log(\hat{Y}_1^{(1)}) + Y_2^{(2)} \log(\hat{Y}_2^{(2)}) + Y_3^{(3)} \log(\hat{Y}_3^{(3)}) \right]$$

$$\hat{Y}_1^{(1)} = \sigma(w_0 + w_1^{(1)} x_{11} + w_2^{(1)} x_{12})$$

$$\hat{Y}_2^{(2)} = \sigma(w_0 + w_1^{(2)} x_{21} + w_2^{(2)} x_{22})$$

$$\hat{Y}_3^{(3)} = \sigma(w_0 + w_1^{(3)} x_{31} + w_2^{(3)} x_{32})$$

$$w_1^{(1)} = w_1^{(1)} - \eta \frac{\partial L}{\partial w_1^{(1)}}$$

$$w_2^{(1)} = w_2^{(1)} - \eta \frac{\partial L}{\partial w_2^{(1)}}$$

enmodel = labelEnmodel()

df['species'] = enmodel.fit_transform(df['species'])